

Euler Integration — Quick Reference

A compact reminder of derivatives, Euler's method, the coffee-cooling example, and the numerical stability condition.

1. Core idea: What is a derivative?

A derivative tells you the **current rate of change** of a quantity. For example, if position x changes with time t , then dx/dt is velocity.

$$dx/dt = \text{current slope} / \text{current rate of change}$$

2. What Euler's method does

Euler looks at the current state, calculates the current slope, and assumes that slope stays approximately constant over the next small time interval dt .

$$x_{\text{new}} = x + dt \cdot dx/dt$$

More generally:

$$x_{n+1} = x_n + dt \cdot f(x_n)$$

Mental model

Look → calculate slope → take a short straight-line step → look again → repeat.

Small dt : the straight-line approximation is usually good. Large dt : the approximation can become inaccurate, and for some systems it can become unstable.

3. Coffee cooling model

Let x be coffee temperature and x_{room} be room temperature.

$$dx/dt = -k(x - x_{\text{room}})$$

Define the temperature error (distance from room temperature):

$$e = x - x_{\text{room}}$$

$$de/dt = -k e$$

What does k mean?

k is a property of the physical system. It describes how quickly the error tends to disappear. Larger k means faster dynamics. Its units are $1/\text{time}$ (for example, s^{-1}).

4. Euler applied to the coffee

Start with Euler:

$$e_{\text{new}} = e + dt \cdot de/dt$$

Since $de/dt = -ke$:

$$e_{\text{new}} = e - k \cdot dt \cdot e$$

$$e_{\text{new}} = (1 - k \cdot dt) \cdot e$$

5. What is r?

r is not a physical parameter. It is simply a convenient name for the Euler multiplier:

$$r = 1 - k \cdot dt$$

Therefore: $e_{\text{new}} = r \cdot e$

So r tells you exactly what happens to the error in one Euler step: the sign tells you whether it flips sides, and |r| tells you whether its magnitude shrinks or grows.

r	What happens
$0 < r < 1$	Error keeps the same sign and shrinks toward zero.
$r = 0$	Euler lands exactly at equilibrium in one step.
$-1 < r < 0$	Error flips sign, but its magnitude shrinks (decaying oscillation).
$r = -1$	Error flips sign with the same magnitude; no decay.
$r < -1$	Error flips sign and grows; numerical solution explodes.

6. Euler stability condition

For the error to shrink, we need:

$$|r| < 1$$

Substitute $r = 1 - k \cdot dt$:

$$|1 - k \cdot dt| < 1$$

$$0 < k \cdot dt < 2$$

$$dt < 2/k$$

This is the Euler stability limit for the simple system $de/dt = -ke$. It is not a universal stability formula for every differential equation.

7. Worked coffee example

Suppose room temperature is 20°C, coffee starts at 90°C, and $k = 0.1 \text{ s}^{-1}$. The initial error is:

$$e_0 = 90 - 20 = 70^\circ\text{C}$$

Case A: $dt = 1 \text{ s}$

$$r = 1 - (0.1)(1) = 0.9$$

Errors: $70 \rightarrow 63 \rightarrow 56.7 \rightarrow 51.03 \rightarrow \dots$

Stable: the error shrinks without changing sign.

Case B: $dt = 15 \text{ s}$

$$r = 1 - (0.1)(15) = -0.5$$

Errors: $70 \rightarrow -35 \rightarrow 17.5 \rightarrow -8.75 \rightarrow 4.375 \rightarrow \dots$

Stable but inaccurate: Euler overshoots room temperature and alternates sides, but the magnitude shrinks because $|r| = 0.5 < 1$.

Case C: $dt = 25$ s

$$r = 1 - (0.1)(25) = -1.5$$

Errors: $70 \rightarrow -105 \rightarrow 157.5 \rightarrow -236.25 \rightarrow 354.375 \rightarrow \dots$

Unstable: the sign flips AND the magnitude grows because $|r| = 1.5 > 1$.

8. Accuracy vs. stability

Accuracy: How close is the numerical solution to the true physical solution?

Stability: Does the numerical solution remain bounded and behave sensibly as the simulation continues?

A larger dt can first make Euler less accurate while remaining stable. If dt becomes too large, the numerical method can become unstable even when the real physical system is stable.

9. The key equations to remember

Idea	Formula
General ODE	$dx/dt = f(x)$
Euler update	$x_{n+1} = x_n + dt \cdot f(x_n)$
Coffee model	$de/dt = -ke$
Euler coffee update	$e_{n+1} = (1 - k \cdot dt) \cdot e_n$
Euler multiplier	$r = 1 - k \cdot dt$
Repeated steps	$e_n = r^n \cdot e_0$
Stability	$ r < 1$
Coffee stability limit	$dt < 2/k$

One-sentence memory aid: Euler looks at the current slope and takes a short straight-line step; for the coffee system, $r = 1 - k \cdot dt$ determines whether each step shrinks, flips, or explodes the temperature error.